

Lucas Perna-Vella

Data Center Security Agent | Python Specialist | MSc Physics

Montreal, QC

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Experience

GardaWorld

Security Agent for eStruxture Data Centers

Montreal, QC

10/2025 - Present

- Ensured physical security of the facility by performing periodic on-foot surveillance patrols and monitoring live camera feeds.
- Controlled access to the facility by answering intercom calls, operating car gates, and assigning access cards.

GardaWorld

Security Agent for Montreal-Trudeau International Airport

Montreal, QC

05/2025 - 10/2025

- Coordinated the flow of vehicles and pedestrians at crosswalks.
- Promoted good traffic fluidity on the arrivals platform by persuading people to not idle in their vehicles.

University of Toronto / SuperCDMS at SNOLAB

Graduate Student Researcher

Toronto, ON / Sudbury, ON

09/2021 - 06/2023

- Used Python to analyze instrumentation data and compute the calibration parameters of a next-generation cryogenic particle detector.
- Created visualizations of results using Matplotlib.
- Presented weekly results to a research group using PowerPoint.
- Managed a small network of Linux servers and Windows computer systems used in research.
- Operated a cryogenic refrigeration system to cool instruments below 50 millikelvin.

Certifications

CNESST Workplace First Aid

Sécurité XGuard

01/2025

IBM Data Science

IBM Skills Network

10/2023

Skills

Programming

Python, SQL, C++

Software

Git, Linux CLI, Microsoft Office

Interpersonal

Communication of results, working as part of a team

Languages

English, French, Italian (beginner)

Data Science

Machine learning (scikit-learn); cleaning, exploration, and transformation (numpy, pandas);
visualization (matplotlib, seaborn)

Education

University of Toronto

Master of Science (MSc) in Physics

Toronto, ON

09/2020 - 11/2021

McGill University

Bachelor of Science (BSc) in Honours Physics

Montreal, QC

09/2016 - 06/2020

Projects

D&D Monster Assessor

github.com/velzenor/DnDMonsterAssessor

- Developed a machine learning pipeline in Python to predict the Challenge Ratings of monsters in Dungeons & Dragons.
- Extracted features from an official Dungeons & Dragons PDF document and compiled data into a cleaned tabular CSV file.